Reproducibility map for "What Prices Reveal About Hidden Trader Populations, and What Identifying Them Costs"

## Source of every number

The replication package contains numbered Colab notebooks and stand-alone scripts; its README lists them in run order. "Part $N$" below and in table notes refers to notebook $N$ (Part 1: theorem checks and spot tests; Part 2: BTCUSDT futures depth; Parts 4–13: the cross-section, replications, fresh-day tests, second venues, the CFTC test, the certified guard and post-hoc checks). Result identifiers such as `X6` or `U6` name the JSON file that the notebook writes. "Kind" distinguishes closed-form or simulation results (recomputed), results from public market data (real data), and audits.

| Result | Source | | Kind |
|---|---|---|---|
| Complete monotonicity, 892 economies | `L26_complete_monotonicity` | | recomputed |
| $d_2 = -\kappa^2 q \delta^2$ | `P1.7-T1_third_tap` | | recomputed |
| Ledger identity; $T_h$ positive definite | `R2-T1`, `R2-T3` | | recomputed |
| Figure S6 | `make_figs2.py` | | recomputed |
| $\delta_{\min}(3)$, Table S12 recomputed columns, 61.14, 29.94 | `recompute_theory.py` `R_P1_theory_constants.json` | → | recomputed |
| Pulse benchmarks (271, 58, 12, 11 pulses), Figures S7–S1 | `make_figs3.py` `R_P1_pulse_benchmarks.json` | → | recomputed |
| Table S32, Table S33, worked example, passive simulation | `recompute_extra.py` `R_P1_extra.json` | → | recomputed |
| Daily noise (Table S15) | `P1.3-T1_sigma_obs` | | real data |
| Table S30 recomputed columns | same | | recomputed |
| $K$-cohort slopes (Table S34) | `T1_K_cohort_rate` | | recomputed |
| ESPRIT slope −6.15 | `P1.2-T1_esprit_rate` | | recomputed |
| Minimum phase, 1,733 economies; rank 6 of 6 | `T2a`, `T2b` | | recomputed |
| Passive slopes −4.21, −5.93 | `T3_passive_rate` | | recomputed |
| Tagged-flow slope 0.011 | `T4_disclosure` | | recomputed |
| Spectral blindness gains | `T5_spectral_blindness` | | recomputed |
| Noise and capital grid (Table S14) | `P1.3-T1_sigma_obs` | | real data |

| Result | Source | | Kind |
|---|---|---|---|
| Loop gain and placebo (Table S20) | `P1.6-T1_ema_placebo` | | real data |
| Estimator table (Table S21) | `P1.5-T1`, `T2`, `T3` | | recomputed |
| Data integrity (111 files) | `R2-T9_data_integrity` | | audit |
| Tables S11, S12 (certified, feasible, adaptive columns) | `recompute_p1_capital.py` `R_P1_capital_ledger.json` | → | recomputed |
| Table 1; resolution floors; Table S2 | `recompute_p1_protocols.py` `R_P1_protocols.json` | → | recomputed |
| SPRT capital figures | `recompute_p1_sprt.py` `R_P1_sprt.json` | → | recomputed |
| Table S9 ($N^*$, $\sigma_{\text{crit}}$) | `recompute_crb.py` `R_P1_crb.json` | → | recomputed |
| Table S9 (ML columns), Table S30 (estimators) | `recompute_p1_estimators.py` | | recomputed |
| Symmetric multi-asset network (Supplement S12) | `recompute_oct20.py` | | recomputed |
| Proof audit: $K_*$ bound and $d_2$ identity (Theorem 4.1), moment gaps (Theorem S25.1), spectral-blindness example (Theorem S12.1) | `recompute_audit.py` `R_audit_oct2026.json` | → | recomputed |
| Theorem 5.2, Table S35 | `kcohort_upper.py` `R_P1_kcohort_upper.json` | → | recomputed |
| Lem. S17.1, Thm S17.2, Tab. S24 | `general_minimax.py` | | recomputed |
| Theorem 6.2 numbers | `disclosure_value.py` `R_P1_disclosure_value.json` | → | recomputed |
| Tables S18, S19, S22 | Part 8: `V2_spot_panel`, `V3_fee_events` | | real data |
| Table S27, top panel | Part 4: `X9_disclosure_metrics` | | real data |
| Table S27, bottom panel; class-position forecasts | Part 6: `Z6_C_identification`, `Z6_secondary`; `P6_fresh_days.json` (65acda36…) | | real data |
| Table S25 | Part 11: `C3_cot_data`, `C4_cot_disclosure`; `P11_cot.json` (6f1d6f35…) | | real data |

| Result | Source | Kind |
|---|---|---|
| Basis-trade signature; real-data Cochrane–Orcutt | Part 13: `R6`, `R7` | real data (post hoc) |
| Feedback across 30 perpetuals | Part 4: `X3_T6_cross_section` | real data |
| Theorem S9.4 full bound (Table S13), direct minimization | `hardening/rh_bound.py`, `rh_check.py` → `R_P1_rh_explicit.json` | recomputed |
| Minimal dimension at $a = 0$; Toeplitz sections; passive slopes for $K \leq 4$; IV determinant | `hardening/p1_checks.py` → `R_P1_hardening_checks.json` | recomputed |
| Stylized nonlinear and trained online rules (Table S16, Figure S4) | `recompute_p1_learners.py` → `R_P1_learners.json` (Panel A); `validate_trained_learners_hardened.py` → `R_P1_trained_learners_hardened.json` (Panels B–C); specification `P1_trained_learners.json` (3e8bf30c…) | recomputed |
| Disclosure in dollars (Table S10) | `recompute_p1_cat_costs.py` → `R_P1_cat_disclosure_dollars.json` | recomputed |
| Out-of-sample post-shock return test (Supplement S19) | `recompute_prereg_oct21_22nd.py` → `R_P1_disclosure_oos.json`; `P1_disclosure_oos.json` (f0182ce5…) | real data |
| Out-of-sample cohort-state trajectory test (Table S28) | `recompute_prereg_oct21_followup.py` → `R_P1_disclosure_state_oos.json`; `P1_disclosure_state_oos.json` (5136cc6b…) | real data |
| Proposition 6.4, Table S36 | `theory/targeted_intervention.py` → `R_P1_targeted_intervention.json` | recomputed |
| Table S30, column $N^*_{\mathrm{CRB}}$; Table S9, spacing-0.10 three-cohort row | `breakthrough_experiments/results_exp17_minimax_lecam_bou` `results_exp15_cohort_information.json` (canonical run) | recomputed |

| Result | Source | Kind |
|---|---|---|
| Table S13, best feasible probes and $L = 320, 480$ | `R_P1_capital_ledger.json`, `R_P1_cquadL.json` (reproduced by `hardening/rh_quad.py`) | recomputed |
| Table S11, peak margins of the eigenvector probes | `breakthrough_experiments/results_exp14_capital_frontier.json` | recomputed (earlier run) |
| Table S24 | `theory/R_P1_general_minimax.json` | recomputed |
| Table S30 and Table S9, estimator columns | `R_P1_estimators.json`, `R_P1_estimators_crb.json` | recomputed |
| Theorem 3.1, numerical check of the equivalence | `R2-T5_merged_invariance` | recomputed |
| Supplement S19 and Table S29: horizon-matched test on 30 new markets (29 of 29, median 7.3, smallest 2.6) and jackknife-corrected persistence (1.0008) | `thesis/evidence/C2_results.json` (specification `thesis/checks/prereg/C2_margin_vs_identify.json`) | real data |
| Supplement S19: exploratory 48- and 96-lag refit (median 7.5) | `thesis/evidence/pilot3_horizon_matched_results.json` | real data (exploratory) |
| Supplement S19 and Table S29: 105-day follow-up (median ratio 74.9; 9 identified; 10 of 30, 23 required) | `thesis/evidence/E1v2_results.json` | real data |
| Supplement S19 and Table S29: 58-day follow-up (22 of 29; 20 of 29, 22 required) | `thesis/evidence/D1_results.json` | real data |
| Supplement S16, Tables S23, S3 and S29: real-noise injection (slopes $-6.01$, $-9.81$; tagged 256–724) | `p1_injection/injection_test.py`, `p1_injection/results/I1_injection_test.json` (specification `p1_injection/prereg/P1_injection.json`, 596a700d…) | semi-synthetic |
| Supplement S3.7 and Table S4: numerical stress tests | `p1_stress/stress_theorems.py`, `p1_stress/results/S1_stress_tests.json` | numerical check |

| Result | Source | Kind |
|---|---|---|
| Supplement S3.8 and Table S5: probing a running market without resets | `p1_running/running_market.py`, `p1_running/results/R1_running_market.json` | numerical check |
| Theorem 3.1(i), Theorem 5.1 and Proposition 6.1 (machine-checked parts) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1MergedModel.lean`, `Paper1Identification.lean`, `Paper1MomentGap.lean`, `Paper1PoleCount.lean`, `Paper1FiniteIdentification.lean`, `Paper1Tagged.lean`, `Paper1Testing.lean`, `Paper1LowerBound.lean`, `Paper1MatchedPair.lean` | Lean 4 proof |
| Theorem 5.2 (machine-checked, every $K$) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1PronyUpper.lean`, `Paper1PronyMoment.lean`, `Paper1PronyDeconv.lean`, `Paper1UniformInverse.lean` | Lean 4 proof |
| Theorems 3.1(ii), S12.1, 5.4 (zeros), 6.2(ii) (tagged upper bound) and Proposition 6.1 (instrument determinant), machine-checked parts | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1MergedSeparation.lean`, `Paper1Blind.lean`, `Paper1MinPhase.lean`, `Paper1TaggedRates.lean`, `Paper1TaggedIV.lean` | Lean 4 proof |

| Result | Source | Kind |
|---|---|---|
| Theorem 5.5(i) and (iii), sample bounds for passive data (machine-checked) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1PassiveCost.lean`, `Paper1PassiveKCohort.lean`, `Paper1PassiveRecord.lean`, `Paper1GaussianTriangular.lean` | Lean 4 proof |
| Theorem S17.2, both parts (machine-checked) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1GeneralMemoryLower.lean`, `Paper1Jackson.lean`, `Paper1JacksonPoly.lean`, `Paper1GeneralMemoryUpper.lean`, `Paper1GeneralMemoryRate.lean` | Lean 4 proof |
| Proposition 6.4, supervisory separation (machine-checked for two groups; part (i) for every $K$) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1SupervisorySeparation.lean` | Lean 4 proof |
| Theorem 4.1(i) and Lemma S24.4, quantity and capital lower bounds (machine-checked) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1QuantityCapital.lean` | Lean 4 proof |
| Pinsker's inequality for events; Theorems 5.1, S25.1, 4.1(i) and Lemma S24.4 at every error level (machine-checked) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1Pinsker.lean`, `Paper1GeneralLevel.lean` | Lean 4 proof |
| Theorem S3.1, finite-window abstention (machine-checked) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1Abstain.lean` | Lean 4 proof |

| Result | Source | Kind |
|---|---|---|
| Corollary 5.3(i)–(ii), identification with unknown $(a, \kappa)$ (machine-checked) | `research_extension/current_manuscript_lean_20261008/CurrentProofs/Paper1UnknownPrimitives.lean` | Lean 4 proof |
| Supplement S18: post hoc robustness (futures prices, roll weeks, 26- and 52-week blocks) | `p11/robustness_cot.py`, `p11res/results/C4R_cot_robustness.json` | real data (post hoc) |
| Supplement S18 and Table S29: two-root test (2 of 8; smaller root $-0.16$ to 0.21) | `p11/two_root_test.py`, `p11res/results/C6M_cot_two_root.json` (specification `p11/prereg_memory/P11M_two_root.json`, e6633588…) | real data |
| Supplement S18 and Table S29: replication on 12 new markets (H1 12 of 12, H2 11 of 12, H3 3 of 12) | `p11/replication_cot.py`, result `C6R_cot_replication.json` (specification `p11/prereg_replication/P11R_cot_replication.json`, 591e71c9…) | real data |
| Figure 1 (paper in one figure) | `papers/paper1/figs/make_fig_summary.py` (inputs `p1_injection/results/I1_injection_test.json`, `p11res/results/C7_data_equivalence.json`) | registered results and post hoc calculation |
| Supplement S18, Table S26: price data needed to match the class reports (median factor 1,353; post hoc) | `p11/data_equivalence.py` → `p11res/results/C7_data_equivalence.json` (inputs `C4_cot_disclosure.json`, `C6R_extract.json`) | post hoc calculation |

| Result | Source | Kind |
|---|---|---|
| Supplement S19 and Table S29: October 2026 tests N2 (1 of 30) and N3 (5 of 30; 21 of 30; 30 of 30) | `p1_oct/oct_tests.py`, `p1_oct/results/zip/results/P1O_N2_feedback.json`, `P1O_N3_combined.json` (specification `p1_oct/prereg/P1O_october2026.json`, 72cbc8ba…; power check `P1O_power_check.json`) | real data |
| Supplement S18 and Table S29: class-flow test (3 of 8; $F \geq 10$ in 7; $w \neq 0$ in 6) | `p11/tagged_flow_test.py`, `p11res/results/C5F_cot_tagged_flow.json` (specification `p11/prereg_flow/P11F_cot_tagged_flow.json`, 052a63f2…) | real data |
| Finite-horizon decision appendix: primary decisions, cost–horizon sensitivities (819 rows), hard-pair requirements, assumption failures, cash-flow audit | `studies/finite_horizon_costs_20261029/run.py` → `results/final_full_20261030/` | real data |
| Finite-horizon appendix: independent checks and LP cross-check | `studies/finite_horizon_costs_20261029/validation/` | real data |